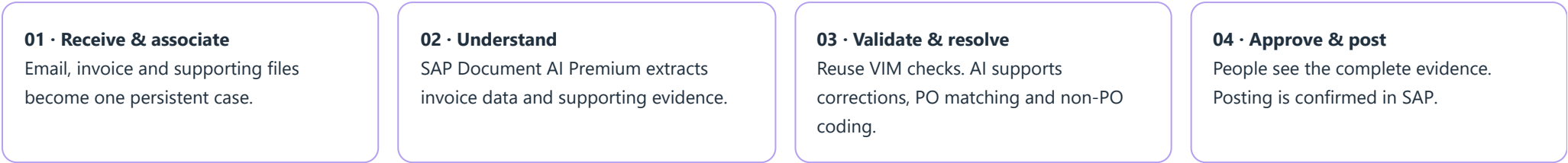


A complete case. From document to business decision.

Proposed architecture · September 2026 · Discussion draft



The components and their responsibilities

Component	Responsibility
New Joule Studio / n8n	Orchestrate tools, reasoning, technical retries and human waits.
SAP Document AI Premium	Extract invoice headers/lines and supporting information through REST APIs.
SAP AI Core Generative AI Hub	Optional classification, evidence interpretation, resolution and coding suggestions.
Custom MCP tools · Cloud Foundry or BTP Kyma	Case operations, extraction adapter, SAP context and supported VIM actions.
Case DB / PostgreSQL + file repository	Documents, relationships, versions, decisions and reviewed resolution history.
VIM / SAP S/4HANA	Authoritative checks, configured business workflow and accounting.

Scope: APAC, Europe and North America; Japan and China excluded. Approximately 184k incoming records annually (provisional reconciliation). PO coding is determined at PO creation; non-PO coding uses business context and supporting evidence.

All walkthrough data is illustrative. No accuracy or automation-rate claim is made. Full design and sources: [architecture.md](#).

Reuse the core. Make the evidence available.

Three integration alternatives

Approach	When it fits	Consequence
A · Reuse VIM (preferred)	Required intake, status and business-action interfaces exist or can be exposed.	New case and AI layer assists the existing business process.
B · Limited integration	Only some VIM operations are accessible.	Automate supported steps; users apply recommendations in VIM. Retain existing intake if necessary.
C · Custom Sony process	Reuse is unsuitable or Sony elects broader replacement.	Sony owns replacement checks, routing, coding and approval; SAP still executes accounting.

Three stories to discuss

- 1. **Non-PO + Excel:** preserve the evidence, resolve a captured-data error, propose a supported allocation, obtain coder and approver decisions, confirm posting.
- 2. **PO matching:** reuse PO account assignment, map ambiguous invoice lines, validate against SAP context and route genuine discrepancies.
- 3. **Late supporting document:** retain a waiting case, associate the later email, revalidate affected information and resume the existing process.

Assumptions to retain

S/4HANA is the target ERP. Sony’s VIM version and usable integration contracts are unconfirmed. New Joule Studio/n8n availability is a deployment dependency. Extraction results require validation. Supporting files can be shared across cases, with versioned associations. Human approval and posting order follow the applicable business process. Document AI performs extraction; the surrounding intake service preserves email/file relationships. Joule Studio’s SAP-managed runtime is distinct from the Cloud Foundry/Kyma deployment of custom extensions. Payment runs and post-posting RTR reallocations remain outside scope.

Sources: customer answers and VIM workbook; supplied SAP Document AI documentation dated 10 September 2026; internal Joule Studio architecture and n8n decks. See architecture.md for references and caveats.